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Shared or Specific? Evidence-Adaptive Sparse Fusion of Graph and Sequence Views for Click-Through Rate Prediction

Anonymous Author(s)

Abstract

Modern click-through rate (CTR) models draw on two complementary sources of evidence: the relational structure linking users, items, creators and categories, and the user's recent behaviour sequence. Graph encoders capture the former and sequence transformers the latter, but the two views are almost always fused by concatenation followed by an MLP. We argue that this is the weak link. The views overlap substantially, so concatenation counts shared evidence twice. It also lets the stronger sequence view dominate exactly where the structural view is most valuable: long-tail items with little or no interaction history. We propose HUG, which keeps the two views separate through encoding and then fuses them by decomposing each into sparse codes over a *shared* dictionary and *view-private* dictionaries. The number of active codes adapts to how much evidence each prediction rests on, so sparse-evidence cases fall back to shared, prior-like codes. Under a strictly causal evaluation protocol on three datasets (short video, news and Q&A), HUG improves AUC by $[x.xx\%]$ over the strongest tuned baseline. The gains concentrate on cold items ($[+x.xx\%]$ AUC in the lowest-evidence bucket). The learned decomposition also quantifies how much information the two views share ($[xx\%]$). We further show that two common shortcuts, frozen graph encoders and item statistics aggregated over the whole window, change conclusions: **[frozen graph encoders underperform a features-only control by $x.xx$ AUC]**.

CCS Concepts

• Information systems → Recommender systems.

Keywords

CTR prediction, graph neural networks, sequential recommendation, multi-view fusion, sparse representations

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1 Introduction

Predicting whether a user will click an item is the core of web-scale recommendation. Two kinds of evidence dominate modern CTR models. The first is *relational*. Users interact with items; items are

made by creators and tagged with categories. This structure is slow-changing, and graph encoders in the LightGCN family exploit it well [16, 48]. The second is *sequential*. The user's most recent actions reveal their current intent, which transformer-based history encoders capture with target attention [5, 45, 49, 54].

Both views are useful, and many systems combine them [4, 40, 51]. The way they are combined has received far less scrutiny than the encoders. The default is to concatenate the view embeddings and let an MLP sort it out. We argue that this default is the weak link, for two reasons.

The views overlap. An item's graph neighbourhood and the sequences it appears in are generated by the same users clicking the same items, so much of what one view encodes the other encodes too. An MLP over the concatenation has no inductive pressure to separate redundant evidence from complementary evidence, and it double-counts the former.

The views are unequally reliable, and the imbalance shifts with the amount of evidence. When a candidate item has a rich history, the sequence view is strong and the structural view adds little. For long-tail items, which make up **[58% of validation impressions on KuaiRand-1K]** **[Note: from the Spec 01 cold bucket; re-measure on the final protocol]**, the sequence view has little to say about the candidate. The structural view is then the main route by which signal reaches the item: through its creator, its categories and co-engaging users. A fusion that weights views the same way for every input cannot express this shift.

These observations suggest a different design principle: *keep the views separate long enough to measure what each contributes, then fuse them in a way that is explicit about shared versus view-specific information and that adapts to the available evidence*. HUG implements this principle. Each view is encoded by a standard, strong encoder: a relation-aware LightGCN with contrastive regularisation for structure, and a target-attention transformer for sequence. The novelty lies in the fusion. Each view's representation is projected onto sparse codes over a dictionary shared by both views plus a private dictionary per view. A cross-view alignment objective makes the shared codes agree; a decorrelation objective keeps the private codes complementary. The number of active codes is governed by an evidence-dependent budget: predictions resting on little evidence are pushed towards a few shared atoms, while well-supported predictions may use many private atoms. The downstream MLP thus receives a representation that is sparse, separates redundant from complementary evidence, and is calibrated to how much the model actually knows.

We deliberately use standard encoders. This isolates the contribution: every fusion variant we compare sees identical encoder outputs, so differences are attributable to fusion alone. The design also yields a quantity that concatenation cannot provide: the fraction of predictive information the two views share. That fraction can be reported, and it can be analysed by evidence level.

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Motivation figure: AUC by candidate-item evidence bucket (0 / 1–4 / ≥ 5 prior interactions) for sequence-only, graph-only, concat fusion, and HUG. Expected shape: concat $\approx$ max(single views) in the cold bucket, while HUG exceeds both.

Figure 1: [Concatenation fusion fails to combine the views where evidence is scarce.] [Note: Populate from heavy-run arms N1, N2, N4 plus the HUG run.]

A second, methodological thread runs through the paper. Graph-based CTR pipelines are unusually easy to leak: interaction edges, item popularity statistics and session structure computed over the full log all contain future information [18]. We evaluate under a strict point-in-time protocol, in which every input for an impression at time t uses only interactions before t . We pair it with a features-only control. We show that this matters: **[under the causal protocol, a frozen (randomly initialised) graph encoder underperforms the features-only control by x.xx AUC, and whole-window item statistics inflate a sequential baseline’s training-time AUC while collapsing its validation AUC].**

Contributions.

- We identify concatenation-based fusion of graph and sequence views as a bottleneck in CTR prediction. We show empirically that it underuses the structural view on long-tail items **[evidence: Fig. 1]**.
- We propose HUG, an evidence-conditioned sparse fusion layer. It encodes the graph and sequence views as sparse codes over a shared dictionary and view-private dictionaries. Each view’s budget of private atoms is set per impression from the interaction evidence available *to that view*, so poorly supported views are forced onto atoms both views agree on (§4.4). To our knowledge, this is the first fusion layer for CTR whose shared/private allocation is tied to per-impression evidence. Because the codes are sparse, the same mechanism yields a countable, per-prediction measure of view overlap.
- We describe and release a causal evaluation protocol for graph-based CTR models: time-masked graph snapshots, point-in-time features, a features-only control, and enforced test-set isolation. We show how common shortcuts change conclusions (§3, RQ5).
- On three datasets (short video, news, Q&A), HUG outperforms tuned sequential, feature-interaction and graph-based CTR baselines **[by x.xx–x.xx AUC]**, with gains concentrated where the mechanism predicts. The learned decomposition reveals **[finding about shared vs. private information across evidence levels]**.

2 Related Work

Graph-based recommendation. Graph neural networks propagate collaborative signal over user–item graphs. LightGCN [16] showed that removing feature transforms and nonlinearities improves collaborative filtering, and contrastive variants (SimGCL, XSimGCL [47, 48]) further help long-tail items. Heterogeneous and knowledge-aware models (R-GCN, KGAT, HGT [17, 33, 38]) model typed relations. Benchmarking studies, however, find that well-tuned simple models are hard to beat [7, 28, 32]. Precomputed, parameter-free propagation (SGC, SIGN [8, 42]) can rival trained message passing. This motivates our frozen-encoder control.

Sequential CTR models. DIN [54] introduced target attention over behaviour histories. BST, SASRec and TransAct [5, 20, 45] use transformers, and HSTU [49] and WuKong [50] study scaling. Feature-interaction models such as DCNv2 and Fi-GNN [22, 37] model cross features without user–item graphs.

Combining graphs and sequences. SR-GNN and GCE-GNN [40, 43] build graphs from sessions. SURGE [4] constructs interest graphs from sequences. GCL4SR, MCLSR and MVCrec [39, 51, 55] align graph and sequence views with contrastive objectives, and Self-GNN [26] combines graph and sequential self-supervision. DCRec [46] unifies sequential patterns with a global collaborative graph through cross-view contrast, using popularity-aware augmentation weights. In CTR, graph-enhanced interest networks target users with sparse behaviour sequences [25]. These works differ mainly in how each view is *encoded* and aligned. Fusion is usually concatenation, attention or a learned gate. Our focus is the fusion layer itself: what is shared, what is view-specific, and how the allocation between the two should depend on each view’s evidence.

Shared/private decomposition. Separating view-invariant from view-specific factors is established in multimodal learning (MISA [15]; the multi-view information bottleneck [6]) and has been carried into recommendation. CMDL [23] and MRdIB [36] disentangle modalities, DisenCDR [2] and CDRIB [3] separate domain-shared from domain-specific user factors, and RIDRec [44] decomposes the predictive roles of semantic and collaborative signals into shared and unique parts, arguing against concatenation. Shared and private *sparse dictionaries* date back to factorised latent spaces with structured sparsity [19]. Recent group-sparse autoencoders study when multimodal dictionaries truly share atoms [21]. HUG adopts this decomposition with sparse codes, applies it to the graph and sequence views of a CTR model, and, unlike prior work, makes the shared/private allocation depend on evidence.

Evidence-adaptive capacity. Tying representation capacity to popularity is a mature idea in recommendation. Mixed-dimension embeddings [11], AutoEmb [53] and PEP [24] size embeddings by frequency or learned pruning. VBAE [58] fuses a collaborative and a feature view through a user-dependent information channel whose bandwidth reflects how sufficient the collaborative evidence is. GateSID [56] gates semantic and collaborative signals by item maturity, with stronger alignment for cold items. HUG differs in where and how capacity adapts. Adaptation happens per impression inside the fusion layer, acts *asymmetrically* on private versus shared atoms, and is driven by a separate evidence signal for each view. Throughout,

“evidence” means an interaction-count proxy (§4.4), not the Dirichlet evidence of evidential deep learning. Sparse coding tools [10, 27, 30] and information-bottleneck objectives [1, 34] underpin the mechanism. In recommendation, discrete codes appear mainly as item tokenisers [31], and embedding collapse has been studied at scale [13].

Evaluation rigour. Data leakage in offline recommender evaluation is well documented [18], as are tuning disparities between proposed models and baselines [7, 32, 57]. CTR models with ID embeddings are known to overfit after one epoch [52]. Our protocol extends point-in-time evaluation to graph inputs.

3 Problem Setting and Causal Protocol

Data. Let $\mathcal{U}, \mathcal{V}, \mathcal{A}, \mathcal{C}$ denote users, items, creators (authors) and categories. The log is a time-ordered set of impressions $\mathcal{L} = \{(u, v, t, y, \mathbf{x})\}$ where $y \in \{0, 1\}$ indicates a click and $\mathbf{x}$ holds context (e.g., page tab, hour). The task is to estimate $P(y = 1 \mid u, v, t, \mathcal{H}_{<t})$, where $\mathcal{H}_{<t}$ is the information available strictly before t .

Point-in-time rule. Every input for an impression at time t is computed from interactions with timestamp $< t$. This covers graph edges, item statistics, behaviour histories and ID vocabularies. Model weights are fit on a training period that strictly precedes validation and test, using a global chronological split ([70/10/20]). Ties never straddle a boundary.

Time-masked graph snapshots. We build one heterogeneous graph with relations $\mathcal{R}$: user–item feedback (click, like, dislike, comment, forward), item–creator, item–category and item–item transitions, plus reverse relations. Every behavioural edge carries its timestamp. The timeline is cut at boundaries $b_0 < b_1 < \dots$ (daily, with the split points inserted). An impression in $[b_k, b_{k+1})$ is encoded from snapshot $\mathcal{G}_k$, which contains only edges with time $< b_k$; degree normalisation is computed from $\mathcal{G}_k$ alone. Item popularity statistics are computed per row from strictly earlier interactions.

Controls and test isolation. Every comparison includes a *features-only* control: the same head and inputs with no graph or sequence encoder. Hyperparameters are selected on validation with an equal trial budget for every model. Test is scored once per reported configuration, and the harness enforces and logs this.

4 Method

HUG (*Heterogeneous Unified Graph*) is built on a single heterogeneous, time-stamped graph: users, items, creators and categories, linked by every feedback type and by in-session transitions. This graph is *unified* in the sense that it is the one source of truth for both views. The graph view propagates over its time-masked snapshots, and the sequence view reads behaviour histories from the same timestamped interactions under the same point-in-time rule. The model is deliberately not unified: the views stay separate until fusion. Figure 2 gives an overview. HUG has three stages: (i) a shared input layer, (ii) two view encoders run in parallel, and (iii) evidence-adaptive sparse fusion followed by a prediction head.

4.1 Input layer

Each node receives an initial embedding $\mathbf{h}^{(0)} \in \mathbb{R}^d$ that combines an ID embedding with projected features. ID vocabularies are built from the training period only; IDs seen fewer than m times share an out-of-vocabulary row. Creators and categories are represented by ID embeddings.

4.2 Graph view

We use a relation-aware variant of LightGCN. For layer l ,

$$\mathbf{h}_i^{(l+1)} = \sum_{r \in \mathcal{R}} \pi_r^{(l)} \sum_{j \in \mathcal{N}_r(i)} \frac{\mathbf{h}_j^{(l)}}{\sqrt{|\mathcal{N}_r(i)| |\mathcal{N}_r(j)|}}, \quad (1)$$

with $\pi^{(l)} = \text{softmax}(\mathbf{a}^{(l)})$, where neighbourhoods come from snapshot $\mathcal{G}_k$ and $\mathbf{a}^{(l)}$ are learned relation logits. The graph representation is the layer mean $\mathbf{g}_i = \frac{1}{L+1} \sum_{l=0}^L \mathbf{h}_i^{(l)}$. Following XSimGCL [47], we add a cross-layer contrastive term $\mathcal{L}_{\text{cl}}$ over the users and items in each batch. The graph-view input to fusion is $\mathbf{e}^G = [\mathbf{g}_u \parallel \mathbf{g}_v \parallel \mathbf{g}_v \odot \mathbf{g}_u]$.

4.3 Sequence view

For an impression (u, v, t) , the history $S_u(t)$ is the user’s last N clicked items before t . Each token is the item’s input embedding plus embeddings of recency rank, log-bucketed time gap and a same-session flag. A light transformer encodes the tokens. The candidate item queries the outputs by target attention, giving $\mathbf{e}^S$.

4.4 Evidence-adaptive sparse fusion

[Note: The method is specified (docs/specs/05-sparse-fusion.md) but not yet implemented. Finalise the equations once validation selects between the top- k schedule and hard-concrete gates.]

Per-view evidence. Each view’s reliability depends on the support it has for this particular prediction. We define, from interactions strictly before t ,

$$n^G = \log(1 + c_v) + \log(1 + c_{a(v)}), \quad (2)$$

$$n^S = \log(1 + |S_u(t)|) + \log(1 + c_v), \quad (3)$$

where c_v and $c_{a(v)}$ are the prior interaction counts of the candidate item and of its creator, and $|S_u(t)|$ is the user’s history length. A new video by an active creator thus has strong graph support but weak sequence support. This asymmetry is what lets the structural view carry cold items.

Shared and private dictionaries. Each view’s input is projected to $\mathbf{p}^v = \text{LN}(W^v \mathbf{e}^v) \in \mathbb{R}^{d'}$, $v \in \{G, S\}$. We learn a shared dictionary $D^{\text{sh}} \in \mathbb{R}^{m_s \times d'}$ and private dictionaries $D^G, D^S \in \mathbb{R}^{m_p \times d'}$, with unit-norm atoms. An encoder maps each view to non-negative codes over the shared atoms and its own private atoms, $[\mathbf{a}^{v, \text{sh}} \parallel \mathbf{a}^{v, \text{pr}}] = \text{ReLU}(E^v \mathbf{p}^v + \mathbf{b}^v)$.

Evidence-dependent sparsity. We keep the top k_{sh} shared codes for every impression. The number of private codes kept for view v grows with that view’s evidence:

$$k_{\text{pr}}^v = k_{\text{min}} + \left\lceil (k_{\text{max}} - k_{\text{min}}) \min(1, n^v / n_{\text{ref}}^v) \right\rceil, \quad (4)$$

Architecture figure: input layer $\rightarrow$ [graph view: relation-aware LightGCN on snapshot $\mathcal{G}_k$] and [sequence view: history transformer + target attention] $\rightarrow$ sparse encoders onto shared dictionary D^{sh} and private dictionaries $D^{\text{G}}, D^{\text{S}}$ with evidence-dependent budget $\rightarrow$ MLP head.

Figure 2: Overview of HUG.

where n_{ref}^v is the 95th percentile of n^v over training impressions. Poorly supported views are therefore restricted to a few private atoms and must express themselves through atoms shared with the other view. Read through the information bottleneck [1, 34], the schedule is a per-impression, per-view rate constraint. A soft alternative places hard-concrete gates [27] on the codes, with an expected- L_0 penalty on private atoms weighted by $\lambda^v(n^v) = \lambda_0/(1+n^v)$. [Report whichever validates better; the other goes in the appendix.]

Objectives. Shared codes from the two views should describe the same evidence, and private codes should carry what the other view lacks:

$$\mathcal{L}_{\text{align}} = \text{InfoNCE}(\mathbf{a}^{\text{G,sh}}, \mathbf{a}^{\text{S,sh}}), \quad (5)$$

$$\mathcal{L}_{\text{dec}} = \|\text{Cov}(\mathbf{a}^{\text{G,pr}}, \mathbf{a}^{\text{S,pr}})\|_F^2 / m_p^2, \quad (6)$$

$$\mathcal{L}_{\text{rec}} = \sum_v \|\text{sg}(\mathbf{p}^v) - D^{\text{sh}\top} \mathbf{a}^{v,\text{sh}} - D^{v\top} \mathbf{a}^{v,\text{pr}}\|_2^2. \quad (7)$$

InfoNCE [35] uses in-batch negatives. $\mathcal{L}_{\text{dec}}$ is the linear-kernel case of HSIC [12]. The reconstruction target is detached (sg), so the projections cannot collapse towards whatever is easiest to reconstruct.

Fused representation and head. The head receives $\mathbf{z} = [\frac{1}{2}(\mathbf{a}^{\text{G,sh}} + \mathbf{a}^{\text{S,sh}}) \parallel \mathbf{a}^{\text{G,pr}} \parallel \mathbf{a}^{\text{S,pr}} \parallel \mathbf{x}]$ and outputs a logit through an MLP. The training objective is

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \alpha \mathcal{L}_{\text{cl}} + \beta \mathcal{L}_{\text{align}} + \gamma \mathcal{L}_{\text{dec}} + \delta \mathcal{L}_{\text{rec}}. \quad (8)$$

The shared-mass ratio $\rho = \sum_v \|\mathbf{a}^{v,\text{sh}}\|_1 / \sum_v \|\mathbf{a}^v\|_1$ gives a per-prediction measure of view overlap. We also report the share of shared atoms that both views actually co-activate, which guards against shared atoms degenerating into private ones [21].

4.5 Complexity

[Per-step cost of graph propagation (sparse matrix products over $|\mathcal{E}_k|$ edges), the transformer ($O(N^2d)$ per row) and fusion ($O((m_s + m_g + m_q)d')$ per row); measured wall-clock in Appendix.]

5 Experiments

We address five research questions. **RQ1:** Does HUG outperform strong, equally tuned baselines? **RQ2:** Is the gain attributable to fusion? **RQ3:** Where do the gains come from? **RQ4:** What do the

Table 1: Dataset statistics after preprocessing (global chronological split).

	KuaiRand-1K	MIND	ZhihuRec
Users	[1,000]	[]	[]
Items	[4.37M]	[]	[]
Train / val / test (M)	8.19 / 1.17 / 2.34	[]	[]
Click rate	0.379	[]	[]
Cold-item test share	[]	[]	[]
Cold-user test share	[]	[]	[]

Re-verify every figure from the final preprocessing logs.]

shared and private codes capture? **RQ5:** How much do protocol shortcuts change conclusions?

5.1 Setup

Datasets. KuaiRand-1K [9] is a short-video log from Kuaishou containing 1,000 users and their impressions, with creator, category and multi-type feedback. MIND [41] is a news-recommendation log with impressions, categories, subcategories and Wikidata entities. Articles churn daily, so most test impressions involve items never seen in training. ZhihuRec [14] is a Q&A log with answers linked to authors, questions and topics, in which most test impressions come from users unseen in training. The three datasets cover three domains and both cold-start axes. For MIND and ZhihuRec we sample users at random ([20% and 12.5%]) and keep each sampled user’s full log. ZhihuRec logs click times separately from impression times, so every label-derived input uses only labels *known* before t . Table 1 summarises the splits.

Baselines. Sequential CTR: TransAct [45]. Scaling and feature interaction: WuKong [50], [DCNv2 [37]]. GNN-based CTR: Fi-GNN [22]. All baselines use the FuxiCTR implementations [57] with a pinned model zoo. Every baseline is given the same row inputs as HUG: identical behaviour histories, point-in-time item statistics, context and categorical features. Each model, including HUG, receives [16] tuning trials on validation.

Fusion variants (RQ2). All fusion variants use identical, jointly trained encoders and differ only in fusion: concatenation + MLP; gated fusion; an *evidence-conditioned* gate (GateSID-style [56], the gate sees $n^{\text{G}}, n^{\text{S}}$); mixture-of-experts routing [29]; dense shared/private

Table 2: Test performance (mean $\pm$ std over seeds). Best in bold; [†] significant vs. best baseline ($p < 0.05$).

Model	AUC	AP	LogLoss	nDCG@10
Features-only control	0.45	0.15	0.65	0.15
TransAct	0.45	0.15	0.65	0.15
WuKong	0.45	0.15	0.65	0.15
Fi-GNN	0.45	0.15	0.65	0.15
[DCNv2]	0.45	0.15	0.65	0.15
Sequence view only	0.45	0.15	0.65	0.15
Graph view only	0.45	0.15	0.65	0.15
Both, concat fusion	0.45	0.15	0.65	0.15
HUG	0.45	0.15	0.65	0.15

AUC by candidate-item evidence (0 / 1–4 / ≥ 5) and by user history length (0 / 1–9 / 10–49 / 50), for the best baseline, concat fusion, and HUG.

Figure 3: Gains by evidence level.

decomposition with the same losses (MISA/CMDL-style [15, 23]); HUG with a constant sparsity budget; and full HUG. The evidence-conditioned gate is the key competitor: it tests whether evidence adaptivity alone, without sparse shared/private codes, explains the gains.

Metrics. AUC, AP, LogLoss and per-user nDCG@10. We report means and standard deviations over [3] seeds. Significance uses a per-user paired bootstrap (1,000 resamples).

Implementation. [Embedding size, layers, dictionary sizes, optimiser, batch size, early stopping; hardware (2 $\times$ 95 GB GPUs).] Code and protocol are released at [anonymous link].

5.2 RQ1: Overall performance

[Narrative: HUG vs. best baseline; whether the graph view adds over the sequence-only model; the features-only control as a floor.]

5.3 RQ2: Is the gain due to fusion?

[Table: fusion variants with identical encoders. Key comparisons: concat vs. HUG (total effect of fusion); dense vs. sparse shared/private (effect of sparsity); fixed vs. adaptive budget (effect of evidence adaptivity).]

5.4 RQ3: Where do the gains come from?

[Expected: the largest gains appear in low-evidence buckets, matching the budget mechanism.]

5.5 RQ4: What do shared and private codes capture?

[(a) Shared-mass ratio ρ vs. evidence level: does low evidence shift mass to shared atoms? (b) Overall view overlap: average ρ , compared with CKA between view embeddings. (c) Atom interpretability: top creators and categories per atom. (d) Learned relation weights $\pi^{(l)}$.]

5.6 RQ5: Do protocol shortcuts change conclusions?

[Table: (i) frozen random graph encoder vs. trained vs. features-only; (ii) whole-window vs. point-in-time item statistics for a sequential baseline; (iii) effect on the ranking of methods. Smoke-scale evidence so far suggests that frozen encoders fall below the features-only control; confirm at full scale.]

6 Conclusion

We argued that fusing graph and sequence views is a first-class design problem in CTR prediction, not an afterthought. HUG keeps the views separate through encoding, then fuses them through sparse shared and private codes whose budget adapts to the available evidence. [Summary of gains and the overlap finding.] Our causal protocol and controls show that [protocol finding]. Limitations: [datasets with dense random exposure; online evaluation; dictionary size sensitivity].

Ethical Considerations

This work uses publicly released, anonymised interaction logs under their licences. No new user data were collected. Improved long-tail recommendation may affect creator exposure; we report results by item popularity to make such effects visible.

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A Protocol Details

[Snapshot construction, tie handling, ID vocabulary, holdout diagnostic, test-access logging.]

B Hyperparameters and Tuning Budgets

[Search spaces and selected configurations for every model.]

C Additional Results

[Validation tables, per-seed results, run time and memory.]